

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
To calculate the quantity of heat required to convert a liter of water from 30 °C to 60 °C (part a), we need to multiply the specific heat of water by the mass of water , the temperature difference , and the conversion factor between calories and joules . First , since 1 liter of water is approximately 1000 grams (g), the mass of water in kilograms is 1 kg . So , the heat gained or released by 1 liter of water (1 kg) can be calculated by :

$$1 \text{ (kg)} (1 \text{ cal/g)} (60 \text{ }^{\circ}\text{C} - 30 \text{ }^{\circ}\text{C}) (1000 \text{ g/L}) = 50,000 \text{ cal .}$$

$$\text{Heat required for 1 liter of water} = 50,000 \text{ } \times 4 = 200,000 \text{ J}$$

$$\text{a) } - 28 \text{ Kcal ,}$$

To calculate heat gained by a block of aluminum when heated from 30 °C to 60 °C (part b), multiply specific heat of aluminum by mass of aluminum and temperature change in degrees Celsius ,
Thermal energy = (steel) (. 215 Cal /g o.c. (1 kg) (60 - 30) inches Brown cow "),
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